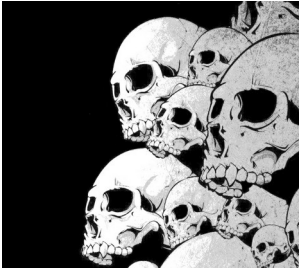


Y. Collette (ycollette.nospam@free.fr)  
<https://audinux.github.io>





# The Live Coding 1/5

SuperCollider : <https://supercollider.github.io>

```
Play {sinosc.ar (onepole.ar (mix (lfsaw.ar ([1,0.99], [0,0.6],  
2000,2000) .trun ([400,600])*[1, -1]), 0.98).
```

<https://www.youtube.com/watch?v=wNWFSIadAH8>

CSound : <https://github.com/csound/csound>

```
sr = 44100  
ksmps = 32  
nchnls = 2  
0dbfs = 1  
  
instr 1  
  
iflg = p4  
asig oscils .7, 220, 0, iflg  
outs asig, asig
```

QuteCsound

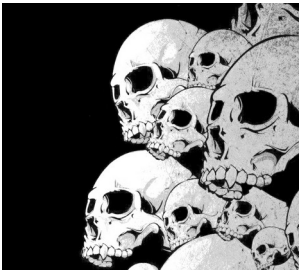
Chuck : <https://chuck.cs.princeton.edu>

```
// set the global gain  
.1 => dac.gain;
```

```
// connect  
SinOsc a => dac;  
110.0 => a.freq;  
1::second => now;  
SinOsc b => dac;  
220.0 => b.freq;
```

miniAudicle

<https://www.youtube.com/watch?v=BHooZu5xzAs>  
<https://www.youtube.com/watch?v=vNrRdyDIniQ>



# The Live Coding 2/5

<https://sonic-pi.net>

Sonic Pi

Run Stop Rec Save Load Size Align Info Help Prefs

```
1 # Rerezzed
2
3 # Coded by Sam Aaron
4
5 use_debug false
6 use_random_seed 103
7 notes = (scale :e1, :minor_pentatonic, num_octaves: 2).shuffle
8
9 live_loop :rerezzed do
10   tick_reset
11   t = 0.02
12   sleep -t
13   with_fx :bitcrusher do
14     s = synth :mod_dsaw, note: :e2, sustain: 8, note_slide: t, release: 0
15     64.times do
16       sleep 0.125
17       control s, note: notes.tick
18     end
19   end
20 end
```

Buffer 0 Buffer 1 Buffer 2 Buffer 3 Buffer 4 Buffer 5 Buffer 6 Buffer 7 Buffer 8 Buffer 9

Préférences

Audio Éditeur Studio Mises à jour

Montre et cache

- ☒ Affichage des numéros de ligne
- ☒ Affichage de la trace
- ☒ Affichage des boutons
- ☒ Affichage des onglets

Look and Feel

- ☐ Mode sombre
- ☐ Plein écran

Automatisation

- ☒ Alignement automatique

Trace

=> Welcome to Sonic Pi

Aide

1 Bienvenue à Sonic Pi

- 1.1 Codage en 'live'
- 1.2 Exploration de l'interface
- 1.3 Apprendre en jouant

2 Synthés

- 2.1 Vos premiers Beeps

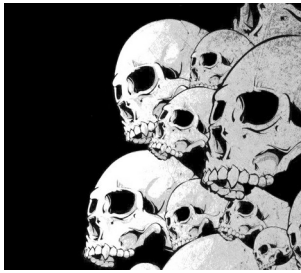
Tutoriel Exemples Synths Fx

Sonic Pi

music\_as :code  
code\_as :art

v2.10-dev

Sonic Pi v2.10.0-dev-24344 on Linux



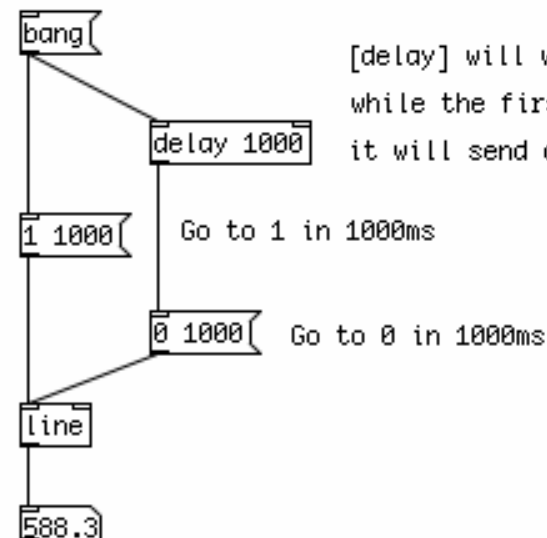
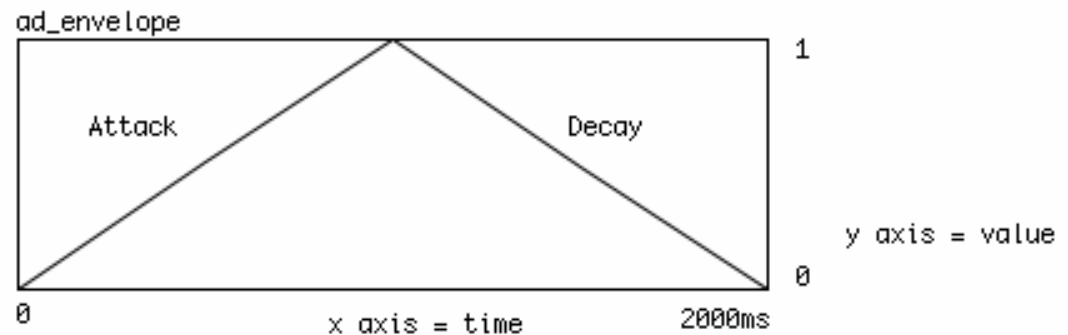
# The Live Coding – 3/5

Pure Data : <https://puredata.info>

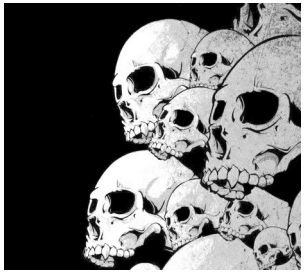
A visual programming tool  
dedicated to audio and  
video.

eg2.pd

Graphical representation of a simple up/down, or  
Attack/Decay (AD) envelope.



[delay] will wait 1000ms after the input "bang",  
while the first ramp is being executed, and then  
it will send a "bang" to trigger the second ramp.



# The Live Coding – 4/5

ProjectM: a video diffuser synchronized at the audio

<https://github.com/projectm-visualizer/projectm>

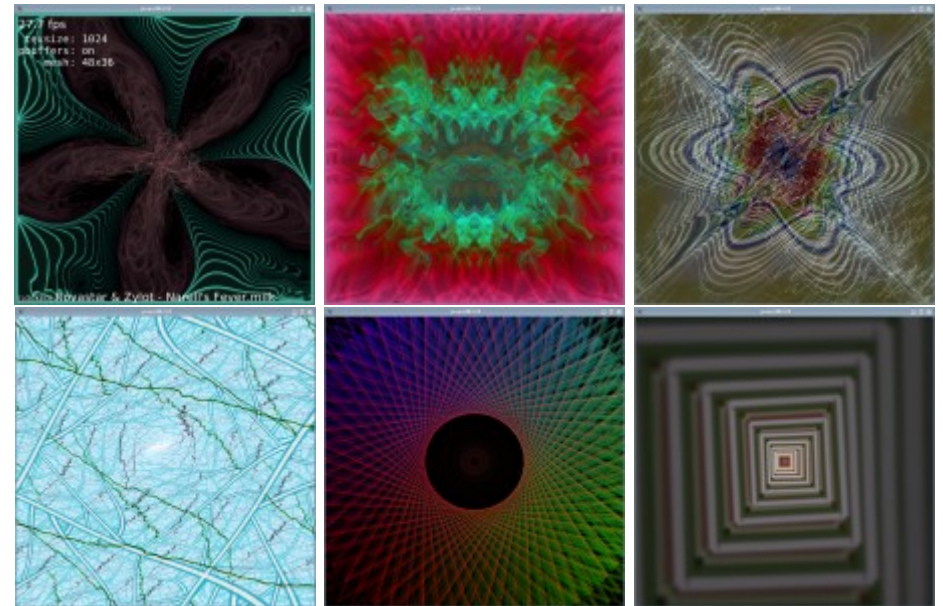
ProjectM code from Winamp.

To launch the Jack version of Projectm

```
$ projectM-jack
```

To launch the Pulseaudio version of Projectm

```
$ projectM-pulseaudio
```



F1: Help

F2: song title

F3: Preset name

F4: Rendering configuration

F5: FPS

F11: Full screen

L: Ventu / delay the preset

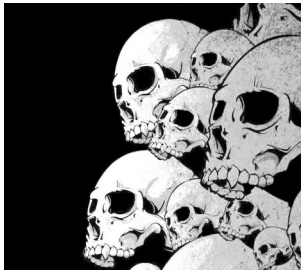
M: displays the menu

A: random preset

N: Next preset

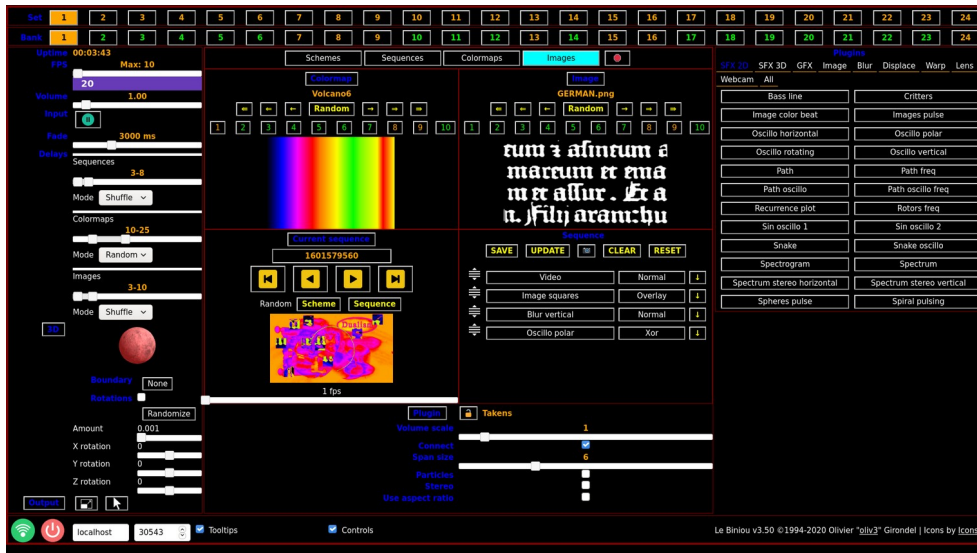
P: Previous preset





# The Live Coding – 5/5

<https://biniou.net>



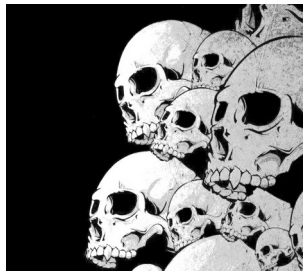
To start Lebiniou:

\$ lebiniou -Input Jackaudio

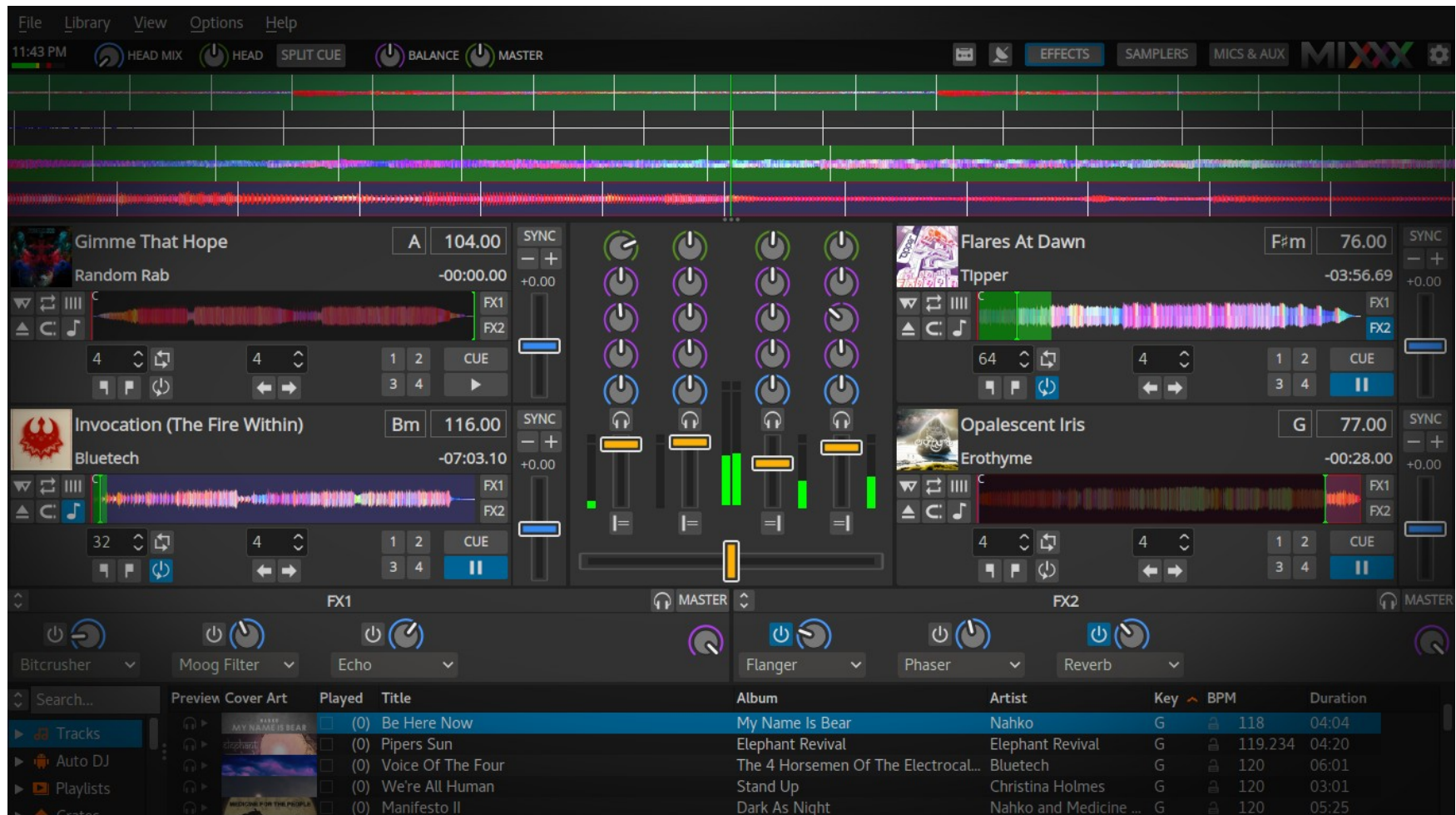
And in the event of a conflict on access to the webcam:

\$ lebiniou -Input Jackaudio -Webcams 0

Lebiniou starts a control window (left) and an animation window (right). You must then connect the lebiniou input to an audio output.



# Mixxx For the Djing



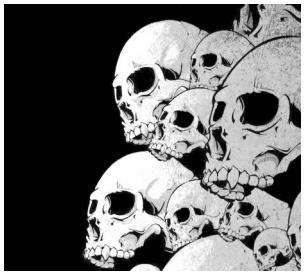
<https://www.mixxx.org>

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Y. Collette

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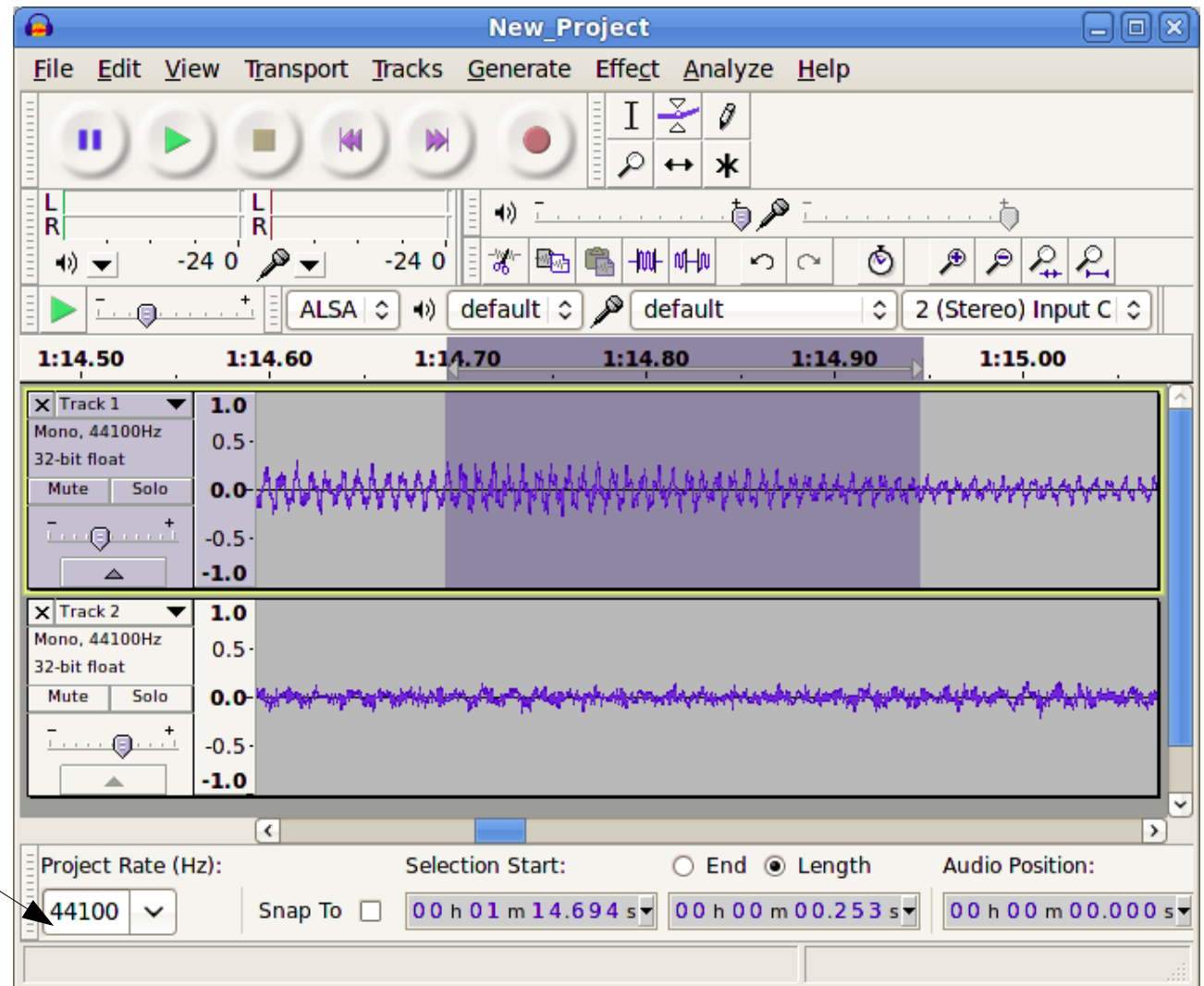
# Audacity

## The audio editor

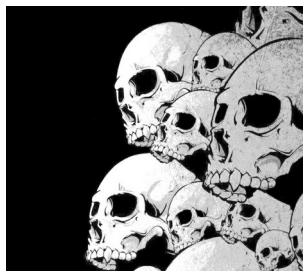
When we use Audacity with Jack, take care to adjust the sampling frequency:

Edition → Preferences → Quality

It will be necessary to match this sampling frequency with that of Jack.

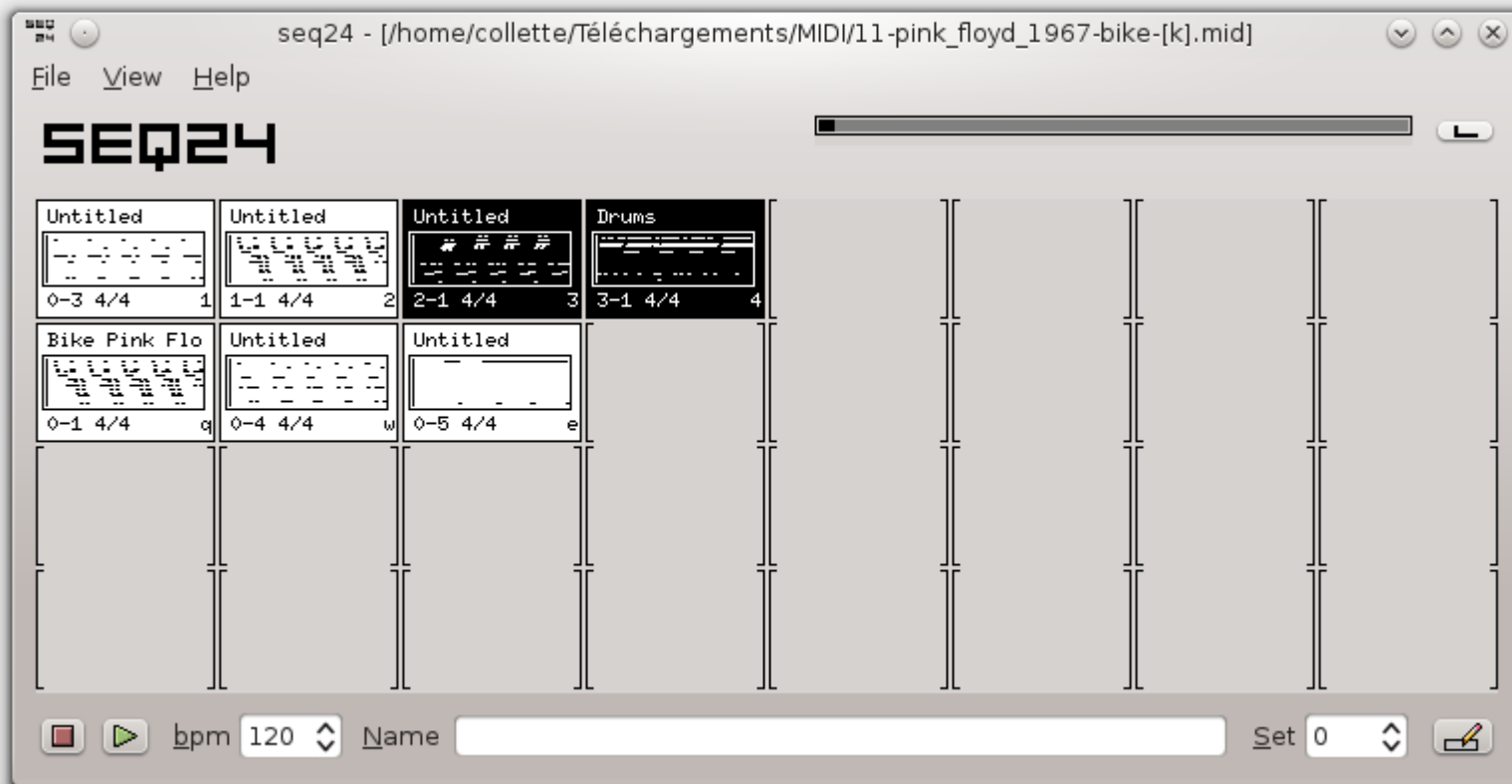




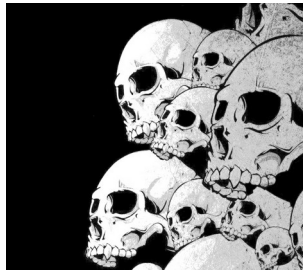


# SEQ24

## A matrix sequencer



<https://launchpad.net/seq24>  
<https://github.com/ahlstromcj/sequencer64>

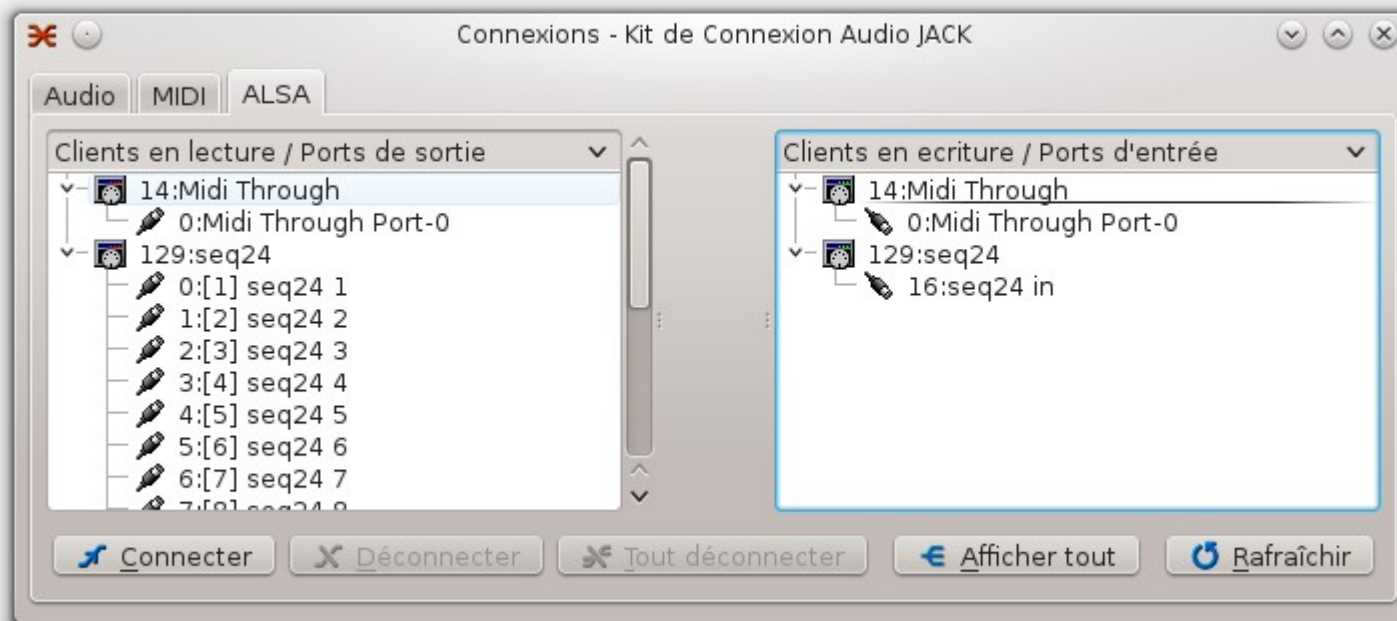


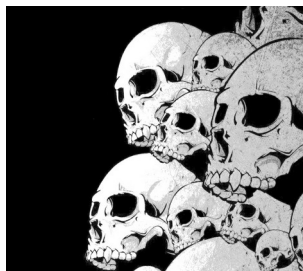
# SEQ24 Jack side

Recommended command line start:

```
$ seq24 -m
```

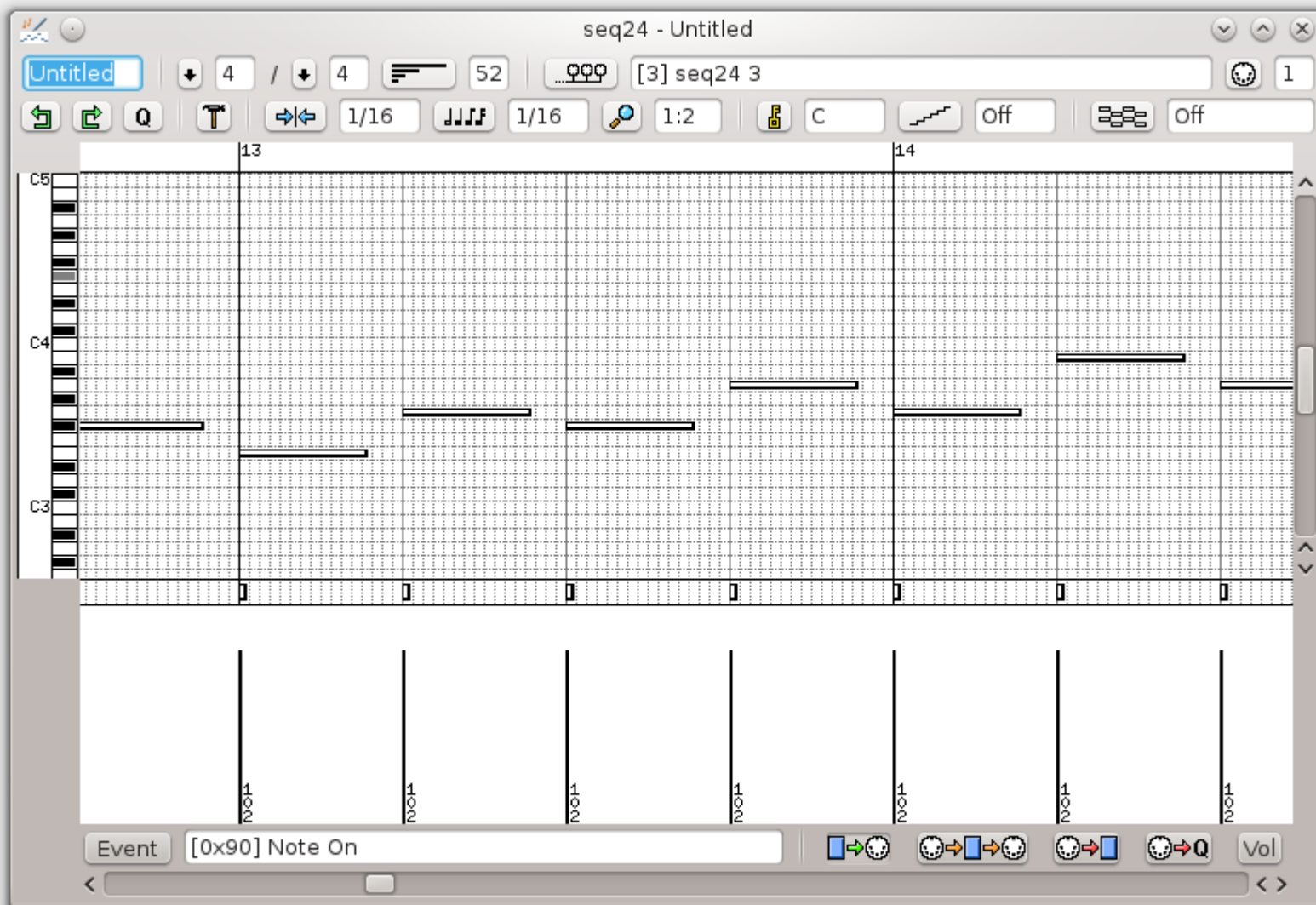
-M, - -Manual\_alsa\_ports: SEQ24 will not reserve Alsa ports



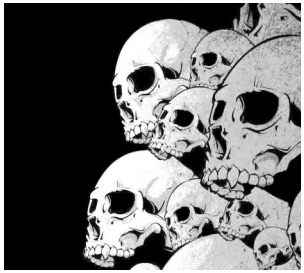


# SEQ24

## The MIDI editor

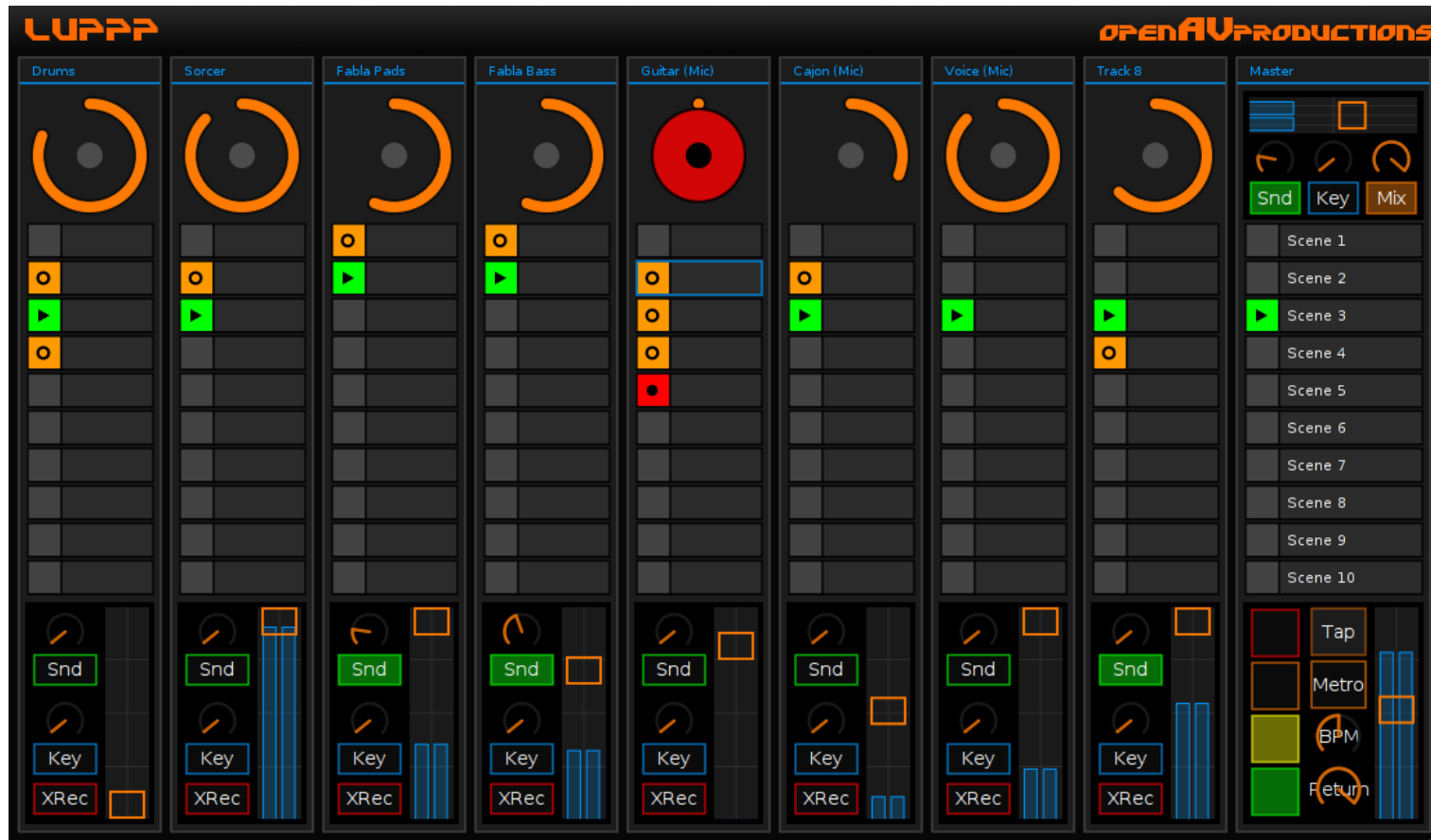


seq192  
seq24  
seq42  
seq66



# OPENAV / LUPPP

## A matrix sequencer



<http://openavproductions.com/luppp>

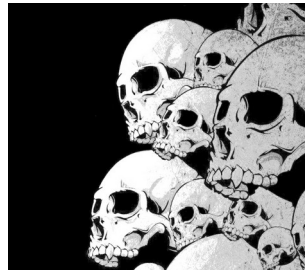


# Improvisor For jazz

The screenshot displays the Impro-Visor software window titled "Impro-Visor: 12-Bar Blues". The interface includes a menu bar (File, Edit, Transpose, View, Play, Utilities, Window, Grammar: My, Preferences, Help) and a toolbar with icons for file operations, playback, and generation. A "Program Status" section at the top right indicates "Click in notes, or type in textual entry field". Below this, a "Textual Entry" field is visible. The main window shows a 12-Bar Blues progression for Clifford Brown, generated from grammars learned from solos of different players. The progression is displayed on a staff with various chords and notes. The chords are: F13 (bar 1), Bb13 (bar 2), Bo7 (bar 3), F13 (bar 4), Cm9 (bar 5), F13b9 (bar 6), Bb13 (bar 7), Bo7 (bar 8), F13 (bar 9), D7#5#9 (bar 10), Gm9 (bar 11), C13b9 (bar 12), F13 (bar 13), D7#5#9 (bar 14), Gm9 (bar 15), and C13b9 (bar 16). The style is set to "swing".

\$ dnf install Impro-Visor

<https://www.cs.hmc.edu/~keller/jazz/improvisor/>



# Improvisor For jazz

To connect improvisor to Qsynth, you must launch the MIDI virtual interface of Alsa:

```
$ sudo modprobe snd-virmidi
```

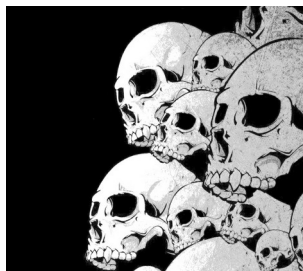
We obtain 4 Virtual Raw Midi as shown the following image:  
In Alsa Out, we have:

- 14: Midi Through
- 20: Virtual RAW MIDI 1-0
- 21: Virtual RAW MIDI 1-1
- 22: Virtual RAW MIDI 1-2
- 23: Virtual RAW MIDI 1-3

After that, just connect improvisor to a virtual Rawmidi and Qsynth input to a virtual Rawmidi output.

In Alsa in, we have:

- 14: Midi Through
- 20: Virtual RAW MIDI 1-0
- 21: Virtual RAW MIDI 1-1
- 22: Virtual RAW MIDI 1-2
- 23: Virtual RAW MIDI 1-3
- 128: Timidity



# Milkytracker



<https://milkytracker.org/>

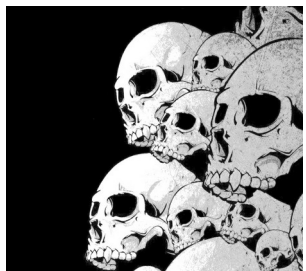


## Historical

Soundtracker – 1987 (Amiga)  
Protracker – 1990 (Amiga)  
Octamed – 1991 (Amiga)  
Scream Tracker 3 – 1993 (PC)  
Fast Tracker 2 – 1995 (PC)  
Impulse Tracker 2 – 1996 (PC)  
Renoise – 2000 (PC & Mac)  
Skalettracker – 2003 (PC)

## File type :

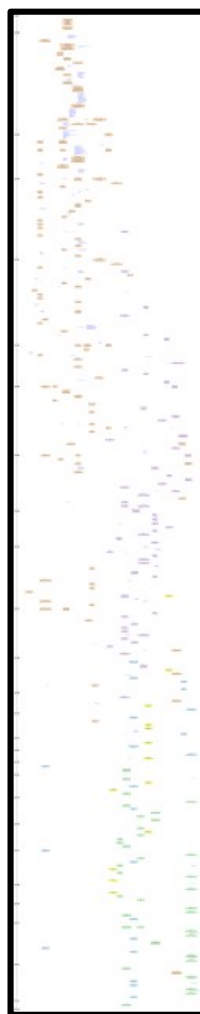
XM - MOD - IT - S3M  
See Wikipedia article



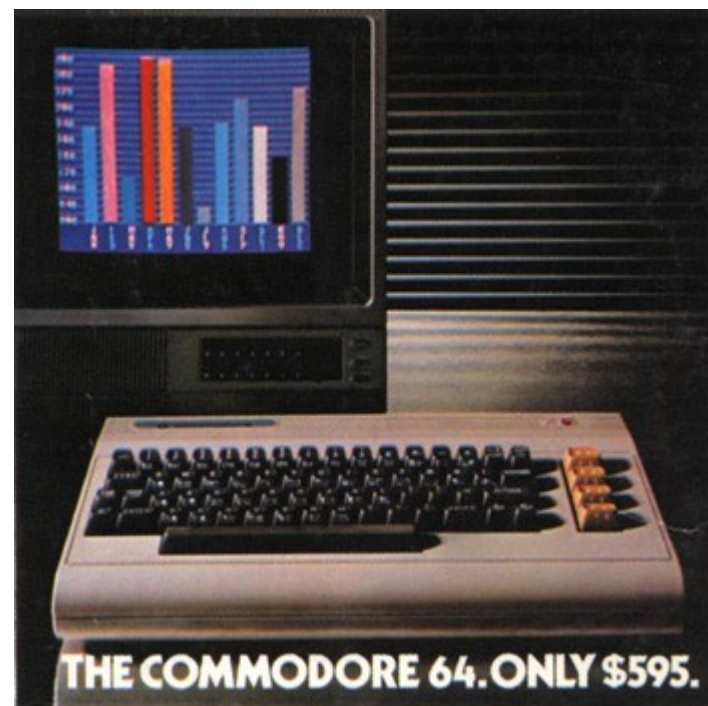
# Milkytracker

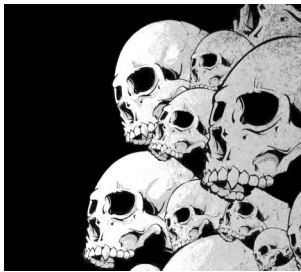
<https://upload.wikimedia.org/wikipedia/commons/1/14/Trackers-historygraph-2012-01.svg>

Amiga - 1987



Commodore - 1982

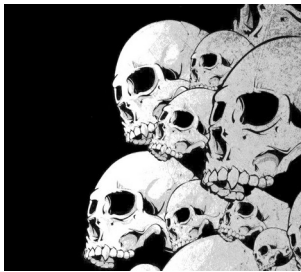




# Klystrack







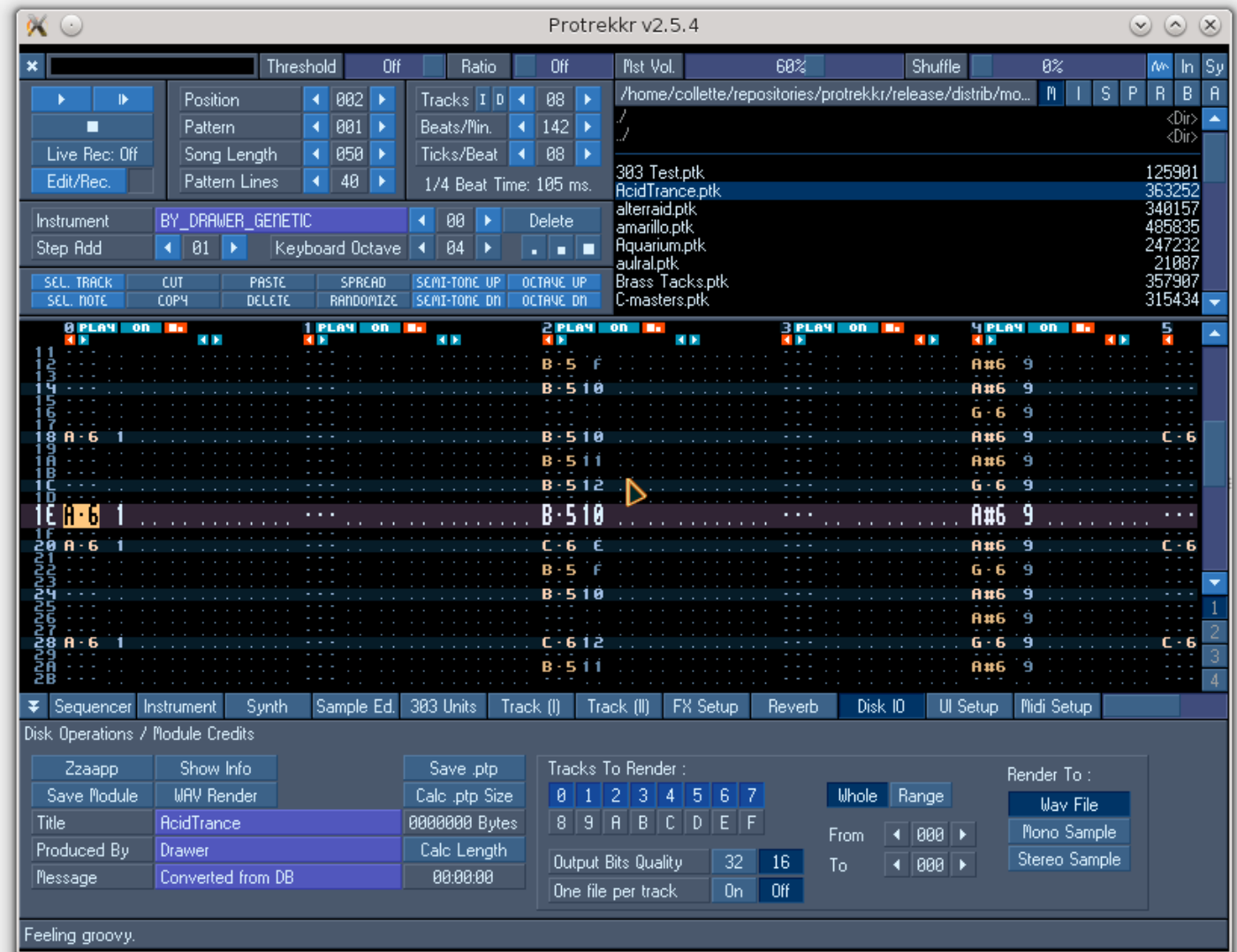
# Protrekkr

<https://github.com/falkTX/protrekkr>

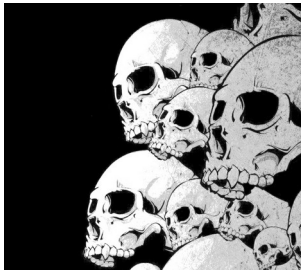
Two versions of Protrekkr exist:

- an OSS version
- a jack version

The version hosted on Github is Jack compatible.

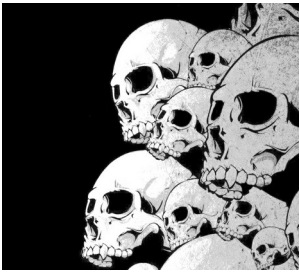


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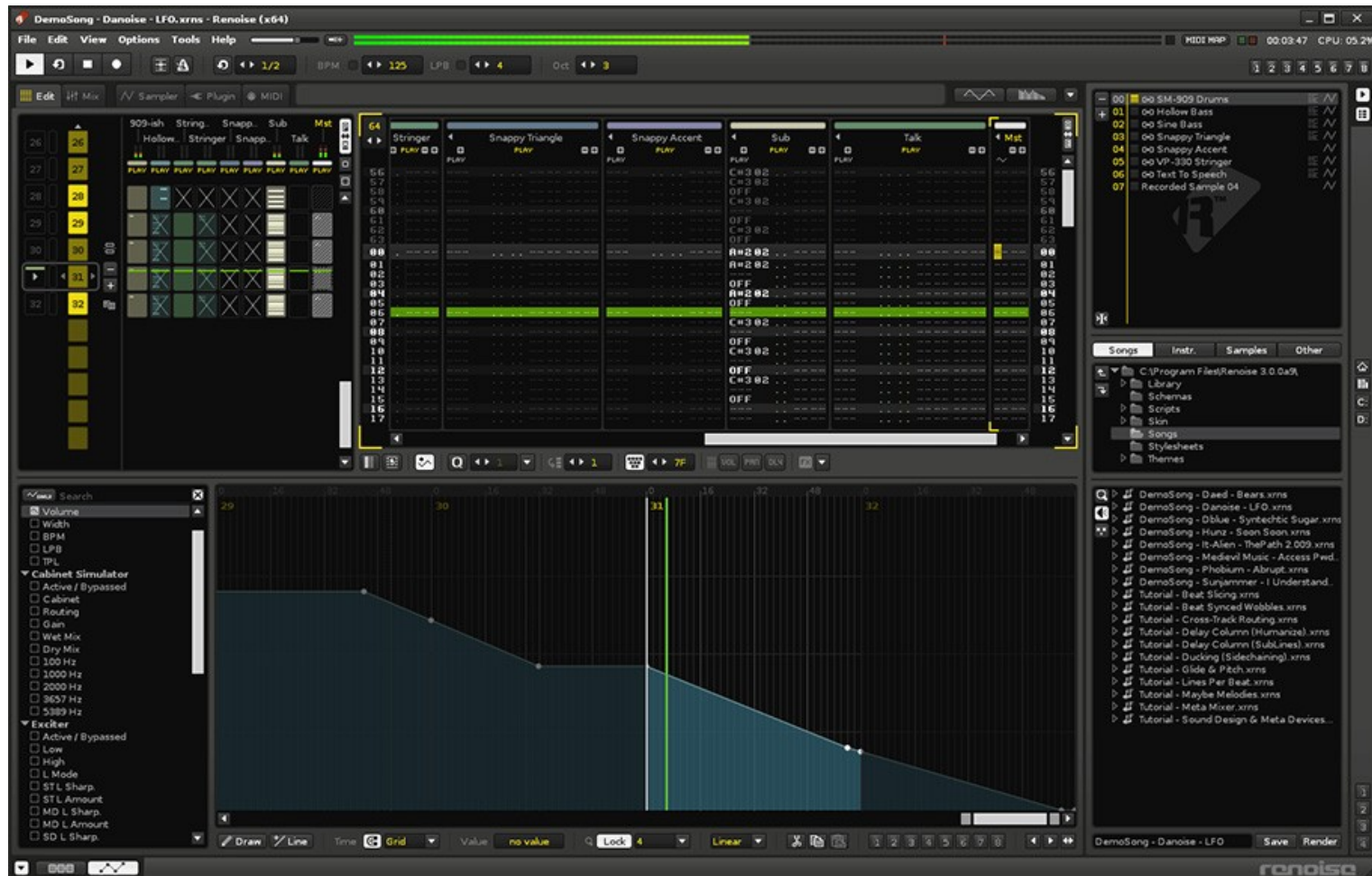
# Trackers

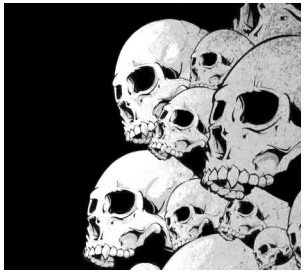
```
$ dnf install BambooTracker  
$ dnf install famitracker  
$ dnf install fasttracker2  
$ dnf install goatracker  
$ dnf install hivelytracker  
$ dnf install plebtracker  
$ dnf install protracker2  
$ dnf install schismtracker  
$ dnf install tiatracker  
$ dnf install soundtracker  
$ dnf install furnace  
$ dnf install protrekkr  
$ dnf install protrekkr2  
$ dnf install tutka  
$ dnf install zytrax
```



# Renoise

<https://www.renoise.com>





# Various

Files for Protrekkr and Milkytracker:

<https://modarchive.org>

Rivendell - Open Source radio

<https://www.rivendellaudio.org>

Jack Net / Jamulus / Ninjam

Music via Internet

<https://jamulus.io>

<https://www.cockos.com/ninjam>



# Webography

Presets of all kinds for Linux tools:

<https://musical-artifacts.com>

Sources of samples:

<https://freesound.org>

<https://archive.org>

[https://wiki.laptop.org/go/free\\_sound\\_samples](https://wiki.laptop.org/go/free_sound_samples)

Documentations of various tools:

<https://en.flossmanuals.net>

Community site:

<http://linuxmao.org/accueil>

<https://librearts.org/>

<https://www.linuxaudio.org>

<https://linuxmusicians.com>

<https://linuxdaw.org/>

Files for mixing:

Nine Inch Nails songs:

<https://nindestruct.com/remix.html>

Different songs:

<https://www.cambridge-mt.com/ms/mtk>

Live Coding resources:

<https://sccode.org/>